Lecture 7 - Social Neuroscience: Reading faces and bodies

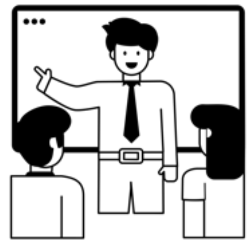
The Social Brain: Critical Perspectives on Science, Society and Neurodiversity

Richard Ramsey

Today

Part 1

- Social Neuroscience:
  Reading faces and bodies



Part 2

- Read articles and discuss

Reading faces and bodies

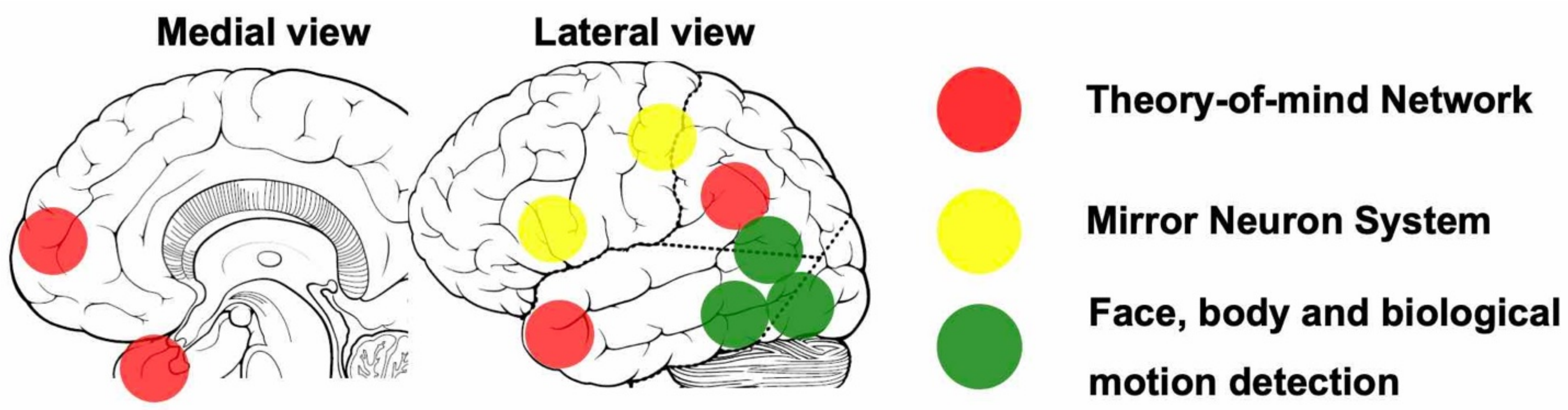
For an overview, See Chapter 5 from Ward (2022)

Overview

- Recap “Social Brain” circuits
- Perceiving faces
- Perceiving bodies
- Joint attention
- Trait inferences from faces and bodies

Recap social brain circuits

Social brain circuits

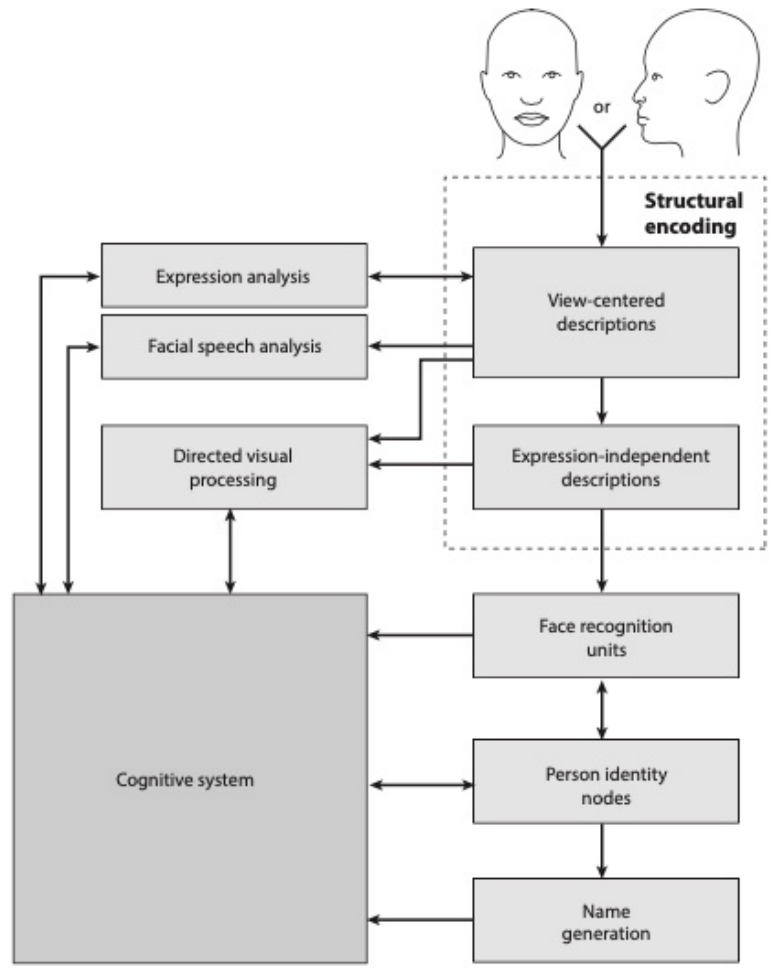


(Adolphs, 2009; Stanley & Adolphs, 2013)

Perceiving faces

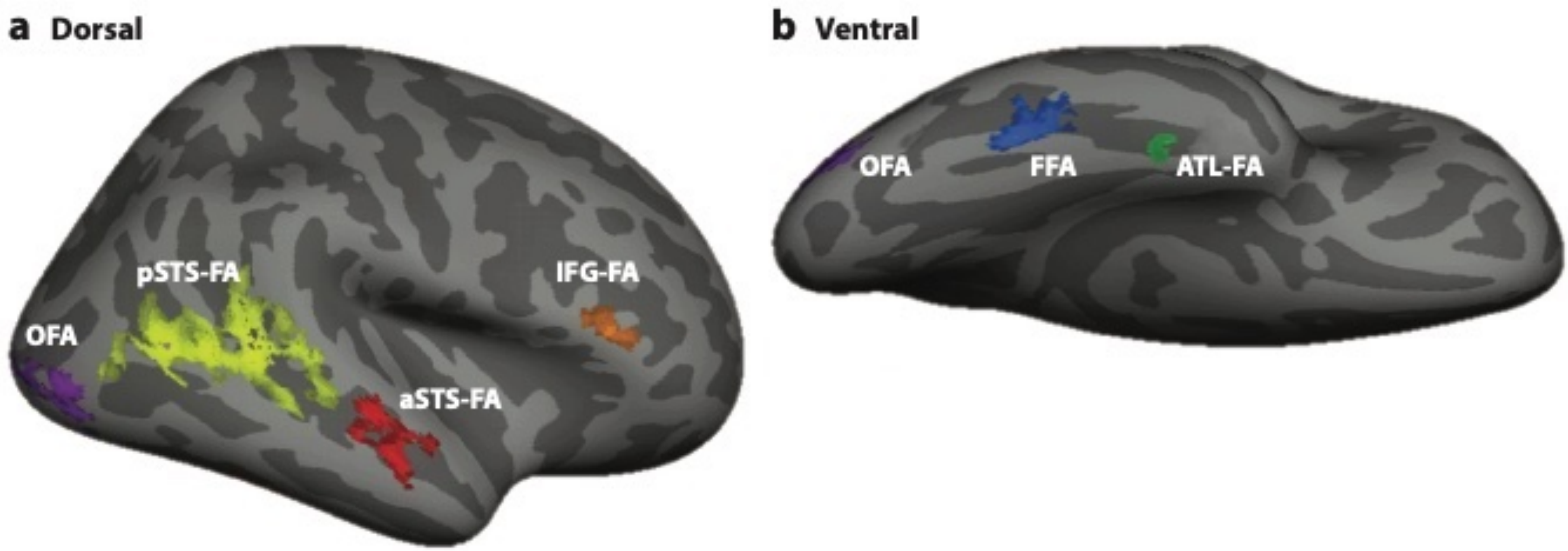
Faces

A cognitive model



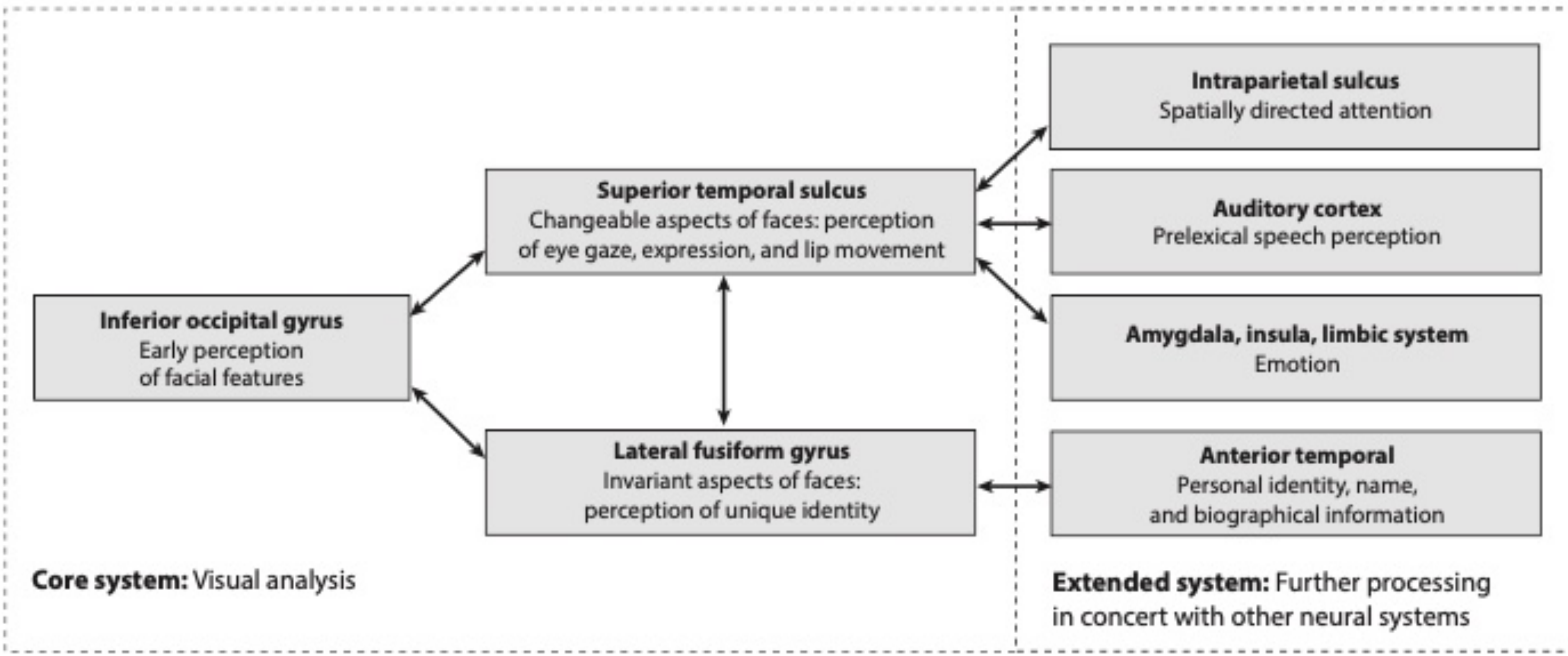
(Bruce & Young, 1986; Duchaine & Yovel, 2015)

Neural basis of face perception



(Duchaine & Yovel, 2015)

Core and extended systems



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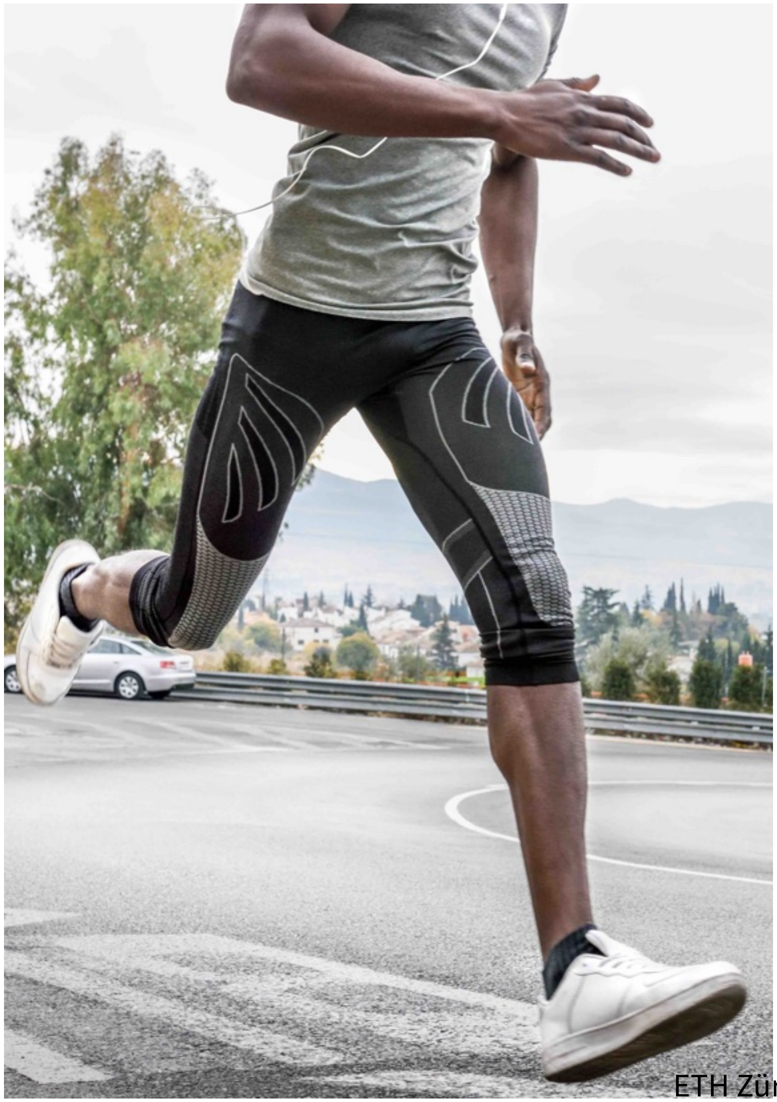
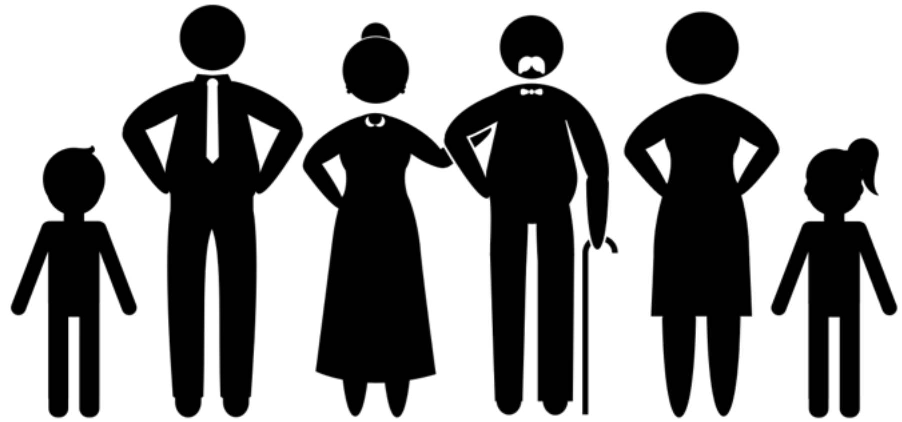
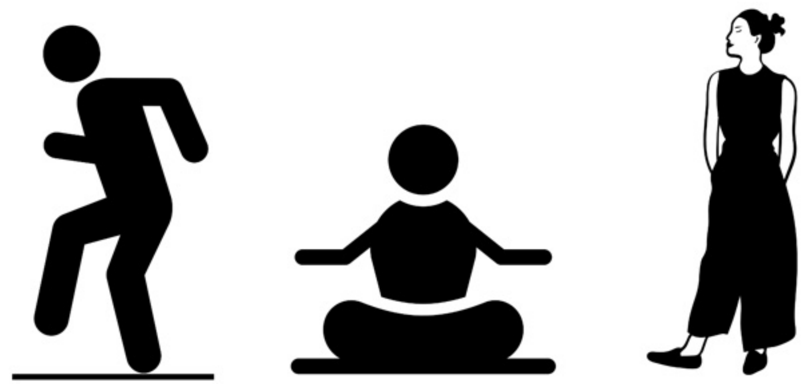
(Duchaine & Yovel, 2015; Haxby et al., 2000)

Perceiving bodies

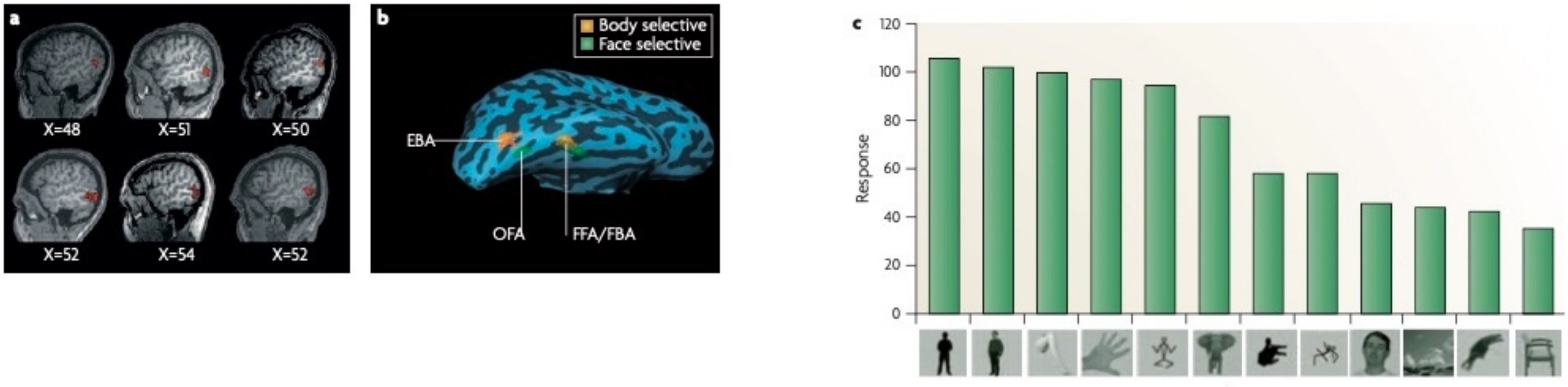
Bodies







Neural basis of body perception



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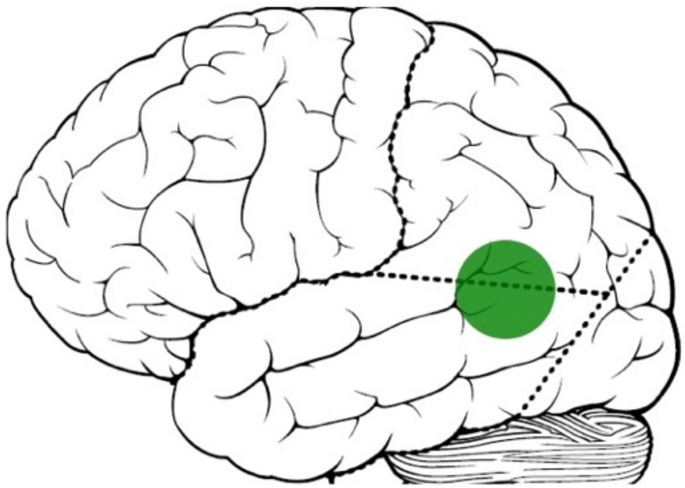
(Peelen & Downing, 2007)

Bodies in motion

Biological Motion - Point Light Display

Neural basis of bodies in motion

Lateral view



posterior superior temporal sulcus (pSTS)

(Grossman et al., 2000; Pitcher & Ungerleider, 2021)

Joint attention

Eye gaze

Fixation　　Cue

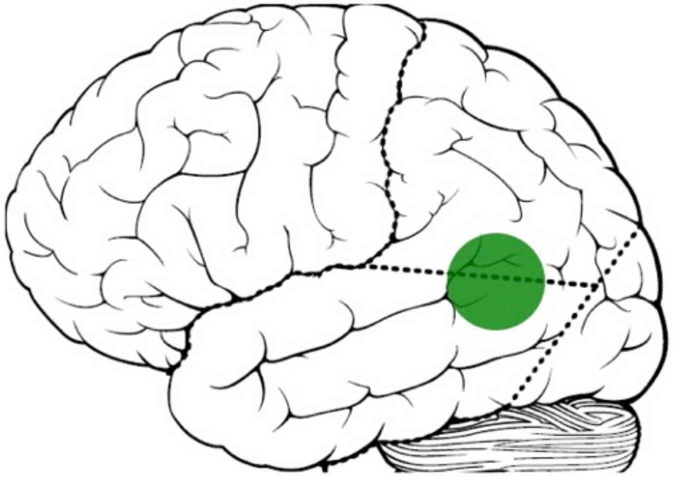
(Frischen et al., 2007)

Eye gaze in social contexts

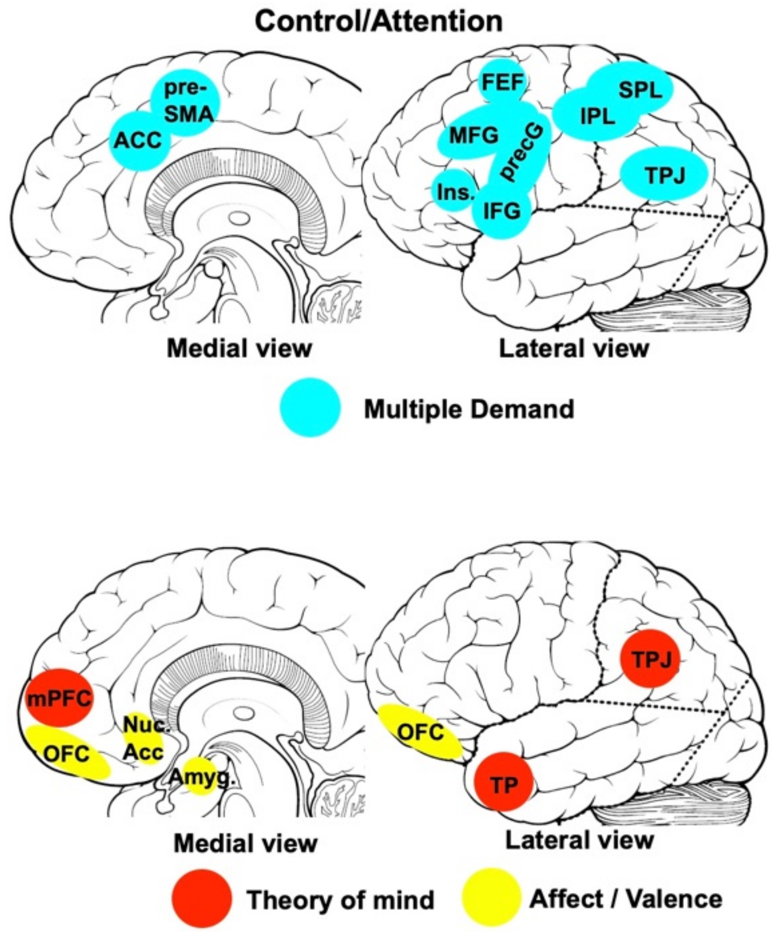
[Le Tricheur à l’as de carreau by Georges de la Tour (c. 1635)]

Neural basis of joint attention

Lateral view



posterior superior temporal sulcus (pSTS)

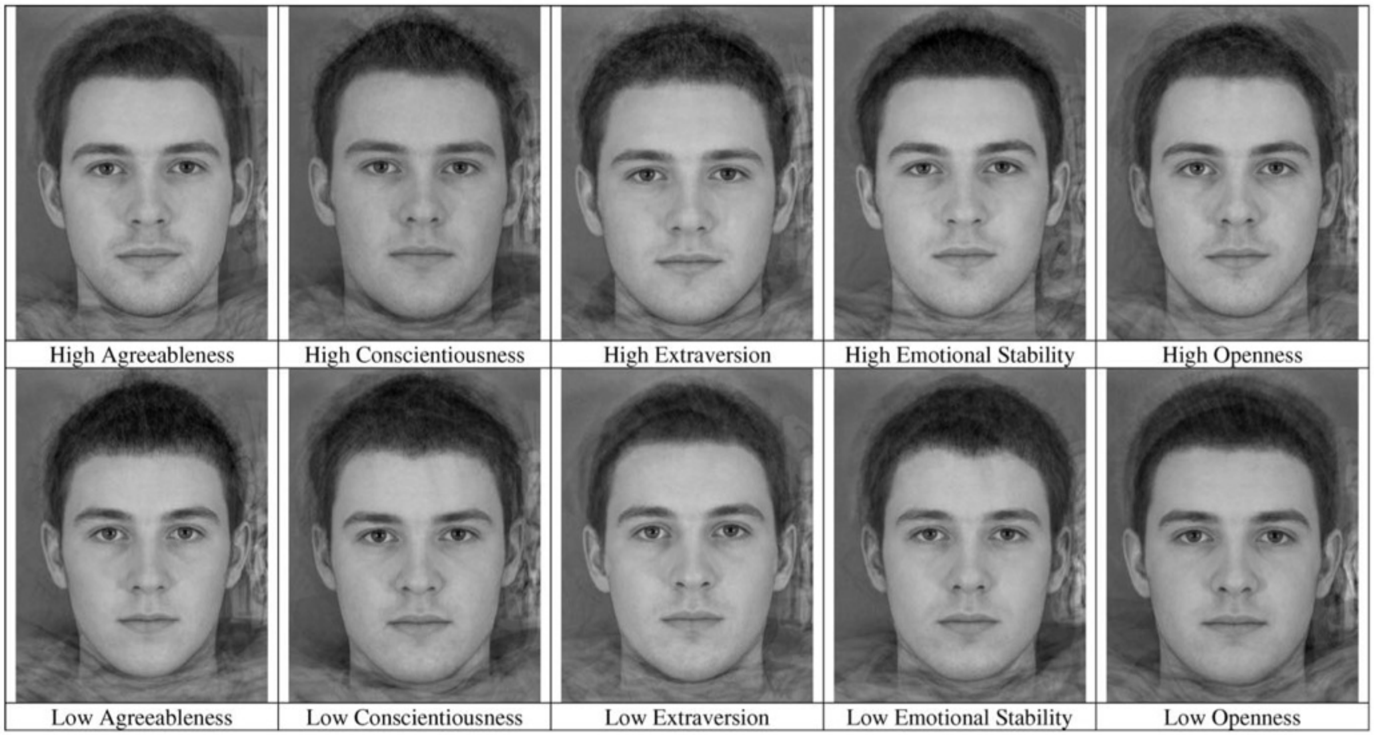


(Hooker et al., 2003; Klein et al., 2009; Redcay & Schilbach, 2019; Shepherd, 2010)

Trait inferences from faces and bodies

Trait inferences from faces

Can we infer personality traits from faces?



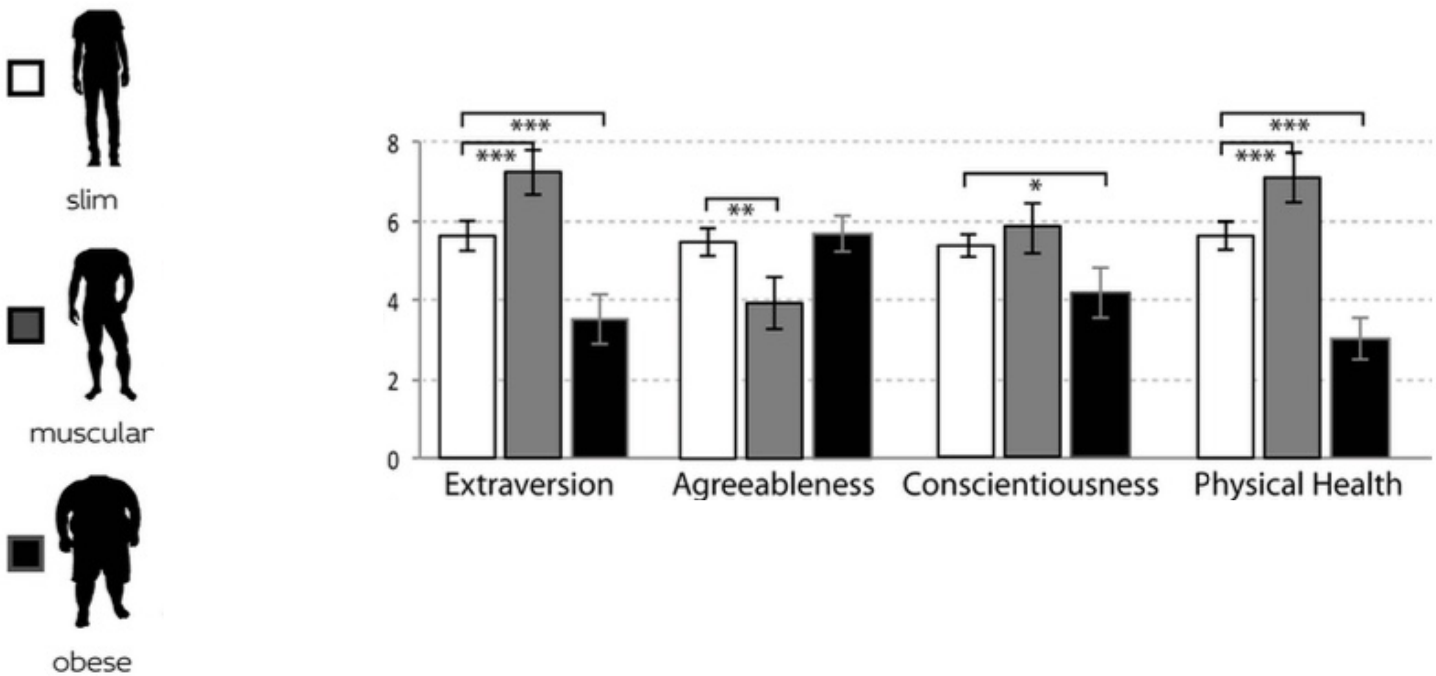
(Penton-Voak et al., 2006; Todorov et al., 2015; Todorov et al., 2025)

Caution

- 'A kernal of truth'
- Self-fulfilling prophecies
- Consensus despite a lack of accuracy

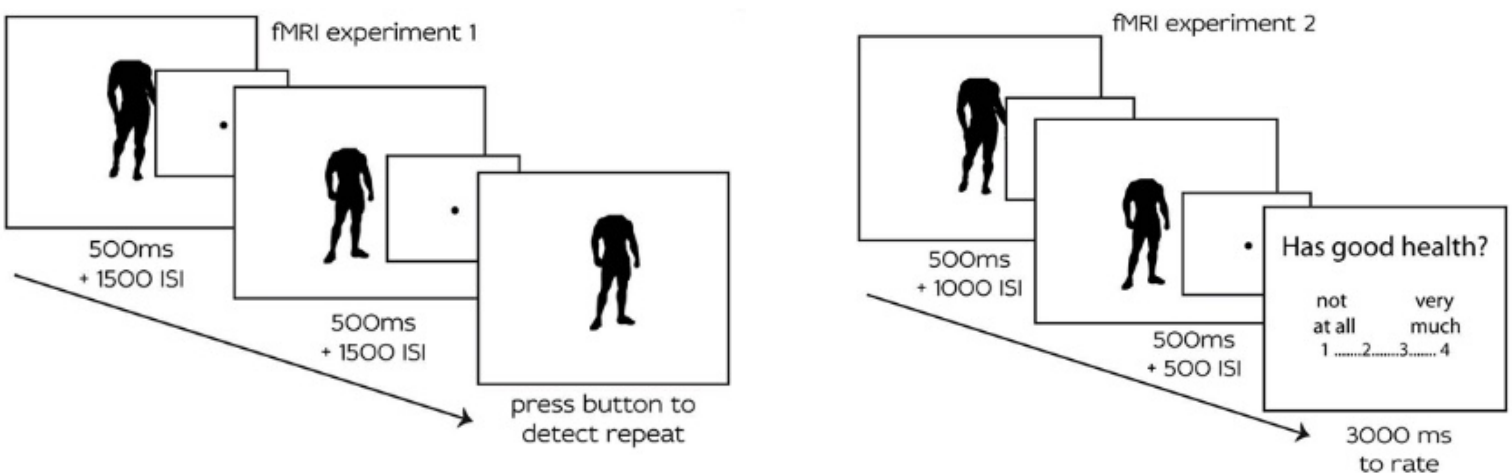
(Todorov et al., 2015; Todorov et al., 2025)

Trait inferences from bodies



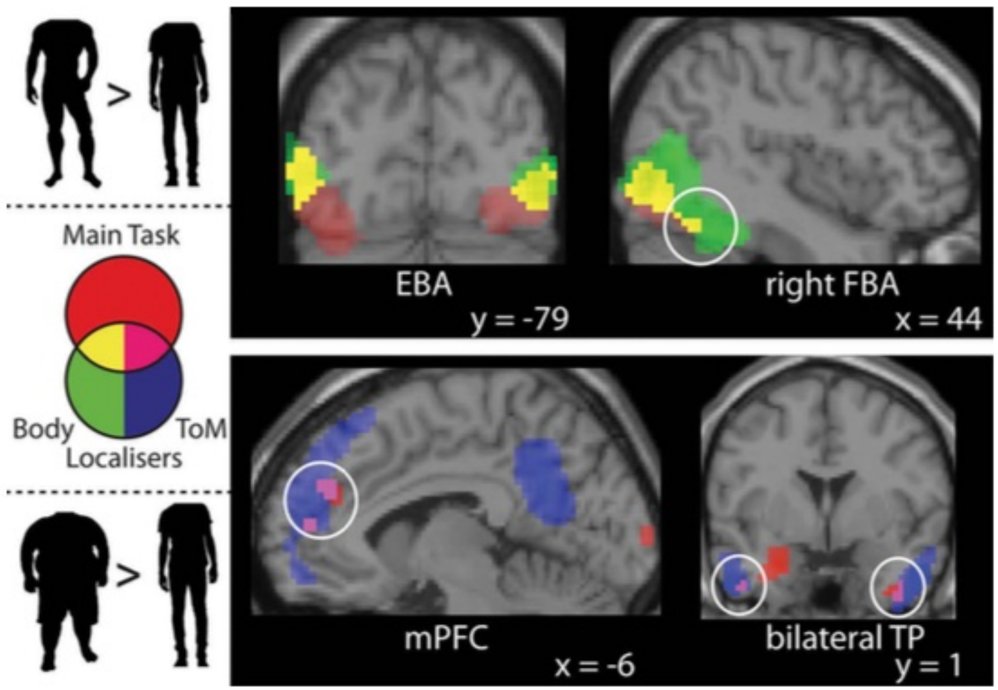
(Greven et al., 2019)

Trait inferences from bodies: fMRI



(Greven et al., 2019)

Trait inferences from bodies: fMRI



- Muscular > Slim engaged body-selective brain areas
- Heavy > Slim engaged the theory-of-mind network
- Only when an explicit judgment was required (Exp. 2 and not Exp. 1)

The Social Brain

(Greven et al., 2019)

Take a break

Part 2 - Read and discuss

Discussion material

- break into small groups (~ 5 per group)
- discuss aspects of the lecture
- discuss aspects of the this article: https://medium.com/@blaisea/physiognomys-new-clothes-f2d4b59fdd6a
- For more details, see also:
- Todorov et al. (2015)
- Chapter 5 from Ward (2022)

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